

Birinder Bachhal

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Education

Boston University

Kilachand Honors College / College of Engineering

Degree: B.S. Mechanical Engineering | Concentrations: Robotics, Manufacturing

Awards/Organizations: QuestBridge Scholar, F1Tenth ACRO, Theta Tau Professional Engineering Fraternity, Society of Asian Scientists and Engineers, Sikh Association at BU

Boston, MA

September 2024 – May 2028

Technical Skills

CAD/CAE: SolidWorks, Fusion 360, AutoCAD, GD&T, Tolerance Stack-Up Analysis, DFM/DFA, FEA, CAM, Technical Drawings

Robotics: Universal Robots (UR12e), RoboDK, Robot Cell Design, End-Effector Design, Offline Programming, Path Planning, LIDAR/Sensor Integration

Manufacturing: 5-Axis CNC Machining, Fixture & Workholding Design, FDM/SLA 3D Printing, Rapid Prototyping, Laser Cutting, Additive Manufacturing

Test & Quality: Metrology, Dimensional Inspection, Calipers, First-Article Inspection, Root Cause Analysis, Design Verification, Bench Testing

Controls/Software: Arduino, Raspberry Pi, PID Control, Stepper Motor Control, Pneumatics, Machine Safety Systems (Light Curtains, E-Stop), Python, MATLAB, JavaScript

Work Experience

The Renatural

Engineering Intern, Robotics & Manufacturing

Brooklyn, NY

July 2026 – Present

- Built a Universal Robots UR12e robotic work cell from scratch, cutting production cycle time from 90 to 45–60 minutes and replacing a manual-intervention 5-axis CNC process
- Designed 22 custom components across a 50-piece structural assembly, integrating Keyence light curtains and E-stop circuits; validated safety interlock function across all access points before production release
- Engineered end-effector mounts for a Zimmer electric gripper and proprietary injection head; simulated and deployed toolpaths via RoboDK executing 50–100 precision arc injections per part
- Designed 3 custom 5-axis CNC workholding fixtures and a pneumatic demolding vessel, applying tolerance stack-up analysis to eliminate part shift; increased usable output 20% and removed operator entry from the machining envelope

Deteqt

Mechanical Engineering Intern

Sydney, Australia

January 2026 – June 2026

- Designed IP65-rated ruggedized enclosure in SolidWorks for 24-hour field deployment above 40°C, delivering GD&T drawings across 4 FDM and SLA prototype iterations verified by dimensional inspection
- Engineered aluminum baseplate heatsink dissipating 2–5W thermal load while preserving quantum sensor performance, applying thermal modeling and DFM for CNC production
- Released a complete production documentation package covering standardized fasteners, heat-set insert specifications, and assembly drawings, enabling outsourced manufacturing with zero engineering follow-up

Projects

BU F1Tenth Autonomous Racing

Vice President (Hardware)

Boston, MA

December 2024 – Present

- Leading mechanical design on a Guinness World Record attempt converting a 2008 Corvette to an autonomous platform
- Redesigned 1:10 scale chassis in SolidWorks, improving structural rigidity 30% and reducing weight 15% through FEA-driven iteration; developed bench test procedures verifying LIDAR and servo communication before track deployment

The Deskinator – Autonomous Workspace Robot

Design Lead

Boston, MA

September 2025 – December 2025

- Designed and machined custom housing integrating NEMA-17 stepper motors and proximity sensors; developed PID control logic on Raspberry Pi across 3 prototype iterations
- Performed root cause analysis on first-article failures, redesigning motor mounts and sensor fixtures to eliminate recurring edge-detection faults

OrthoCAD Pro – Automated Orthotic Design Platform

Personal-Interest Project

Boston, MA

July 2025 – September 2025

- Built a smartphone photogrammetry pipeline in JavaScript achieving ± 1 mm scan repeatability, automating manufacturing-ready CAD generation in Fusion 360 and SolidWorks

First-Year Innovation Fellowship – Closed-Loop Insulin Delivery

2024-2025 Fellow

Boston, MA

September 2024 – June 2025

- Designed 2 functional prototypes integrating mechanical hardware with embedded control logic modeling real-time glucose regulation

Leadership

Vice President: BU F1Tenth Autonomous Racing

December 2024 – Present

Executive Board Member: Sikh Association at Boston University

June 2025 – Present

Academic Chair: Theta Tau Professional Engineering Fraternity

June 2025 – January 2026